Artificial Intelligence in Healthcare

Artificial Intelligence (AI) is transforming healthcare by improving diagnosis, treatment planning, and patient monitoring. Machine learning models analyze medical images such as X-rays and MRIs to help detect diseases earlier. Natural Language Processing (NLP) is used to summarize clinical notes and extract useful information from patient records. AI-powered wearable devices monitor heart rate, blood oxygen levels, sleep, and physical activity. These systems can notify doctors or patients when abnormal health patterns are detected. However, AI adoption also introduces challenges. Hospitals must protect patient privacy, secure medical data, reduce algorithmic bias, and satisfy regulatory requirements before deploying AI systems. In the future, AI is expected to enable personalized medicine by recommending treatments based on genetic, lifestyle, and clinical data. Even with these advances, AI is intended to assist healthcare professionals rather than replace them because final medical decisions require human expertise.